CHAPTER 9: Git & Version Control

Question 1: Hacktoberfest Participation, Commands Used, and Inclusion in Course

Participation & ~~Commands~~ Commands: Creating and sharing repositories involves core Git commands: git init, git clone, git status, git add, git commit -m, git branch, git checkout -b, git remote add origin, git push -u origin main, git pull, and creating Pull Requests on GitHub.

Opinion on Inclusion in S&AI Course: Including Git in ~~tot~~ the first semester is essential. Code that isn't version-controlled is fragile. Learning Git early trains students to work collaboratively, safely experiment with branching without fear of breaking working code, and prepares them for open-source contributions and real industry engineering workflows from day one.

Question 2: Who created Git, when, and why?

Creator & Date: Created by Linus Torvalds (the creator of the Linux kernel) in April 2005.

Origin & Why: The Linux kernel development community had been using a proprietary distributed version control system called BitKeeper. In early 2005, BitKeeper revoked the free access license for the Linux community following reverse-engineering disputes. Frustrated with existing open-source alternatives like CVS and SVN (which were centralized, slow, and handled branching poorly), Linus took a break from kernel maintenance and engineered Git from scratch in a few weeks. He designed it to be fully distributed, lightning fast, resilient to data corruption, and optimized for non-linear development (rapid branching and merging).

Question 3: What does "working tree clean" mean in git status?

Meaning: It means your working directory currently matches the latest commit of your checked-out branch. There are no untracked files, no uncommitted modifications, and no staged changes waiting to be recorded.

Personal experience: Yes, you see it whenever you run git status immediately after finishing a git commit or after ~~cotning~~ cloning a clean repository.